

Pharmacology of Antiarrhythmic Drugs

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Electrical physiology of the heart

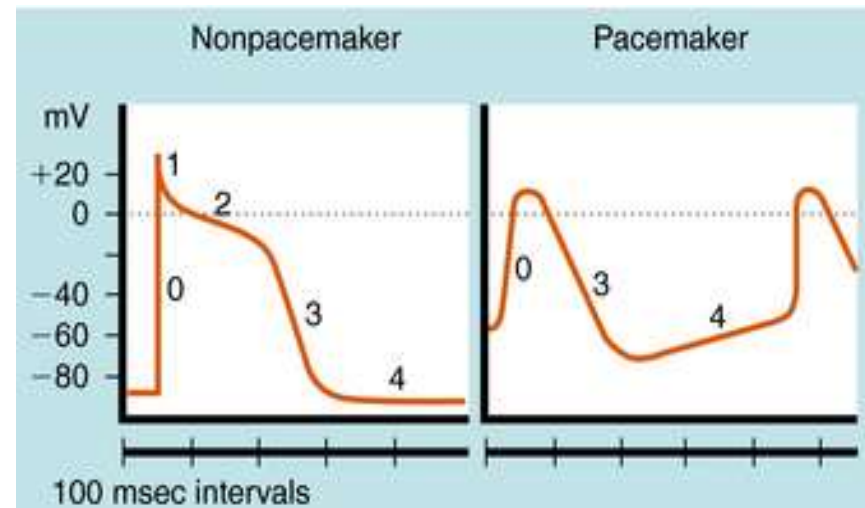
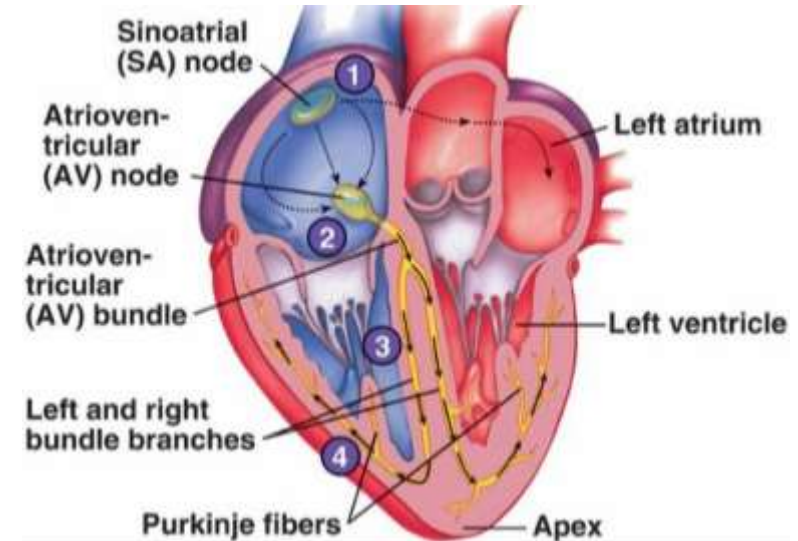
- Human heart is ***both a mechanical and an electrical organ***
 - Mechanical component**: Pumps the blood
 - Electrical component**: Controls the rhythm of the pump
- Two types of cardiac myocytes

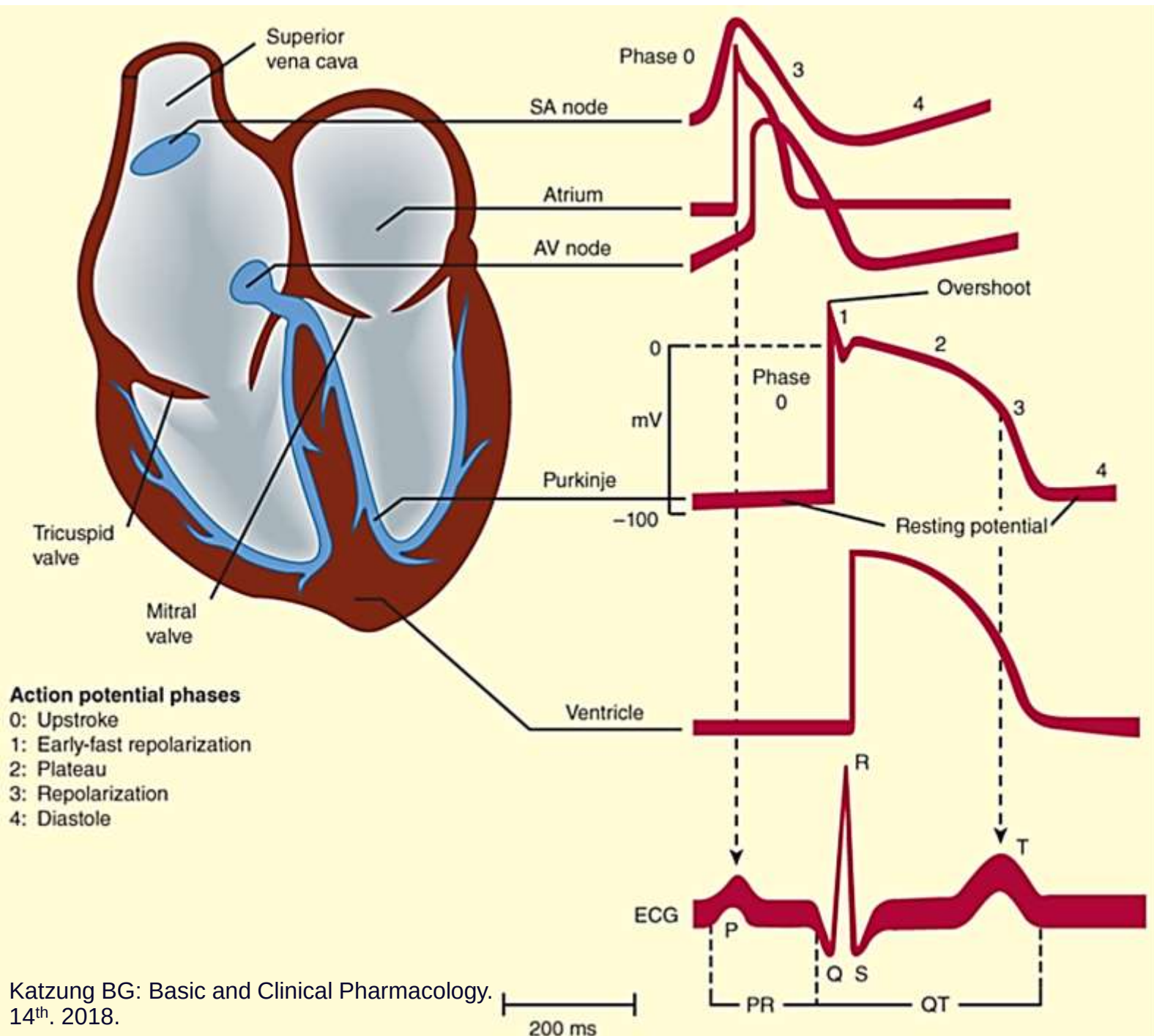
Pacemaker cells

- Automaticity**: Spontaneously initiate action potentials
- Found in **SA node, AV node, and ventricular conducting system** (bundle of His, bundle branches, and Purkinje fibers)

Non-pacemaker cells

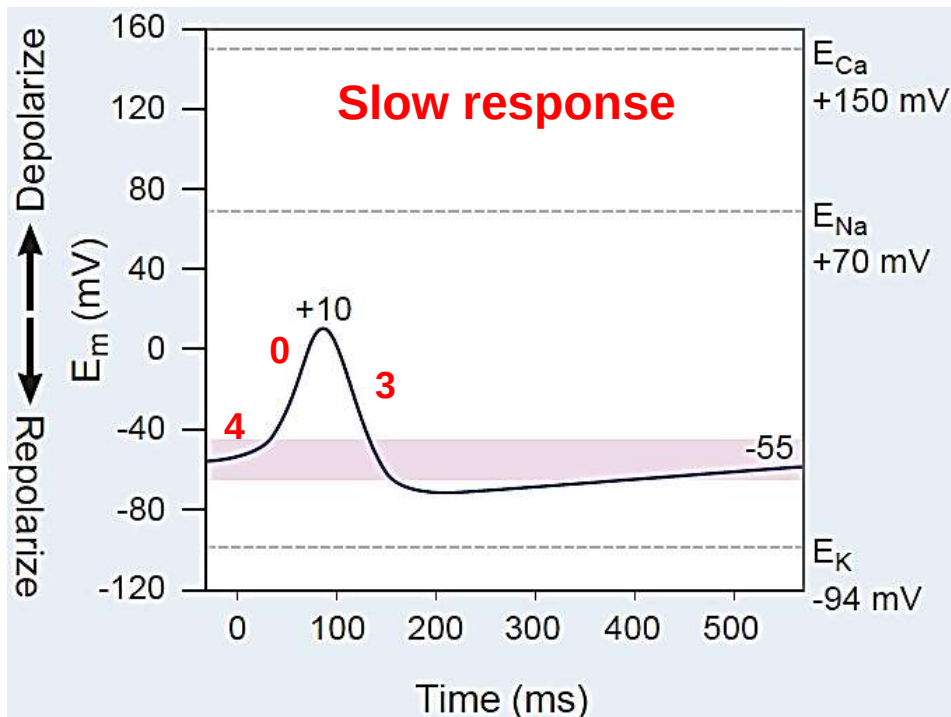
- No automaticity**: Control cardiac contraction
- Found in **atrial and ventricular myocytes**



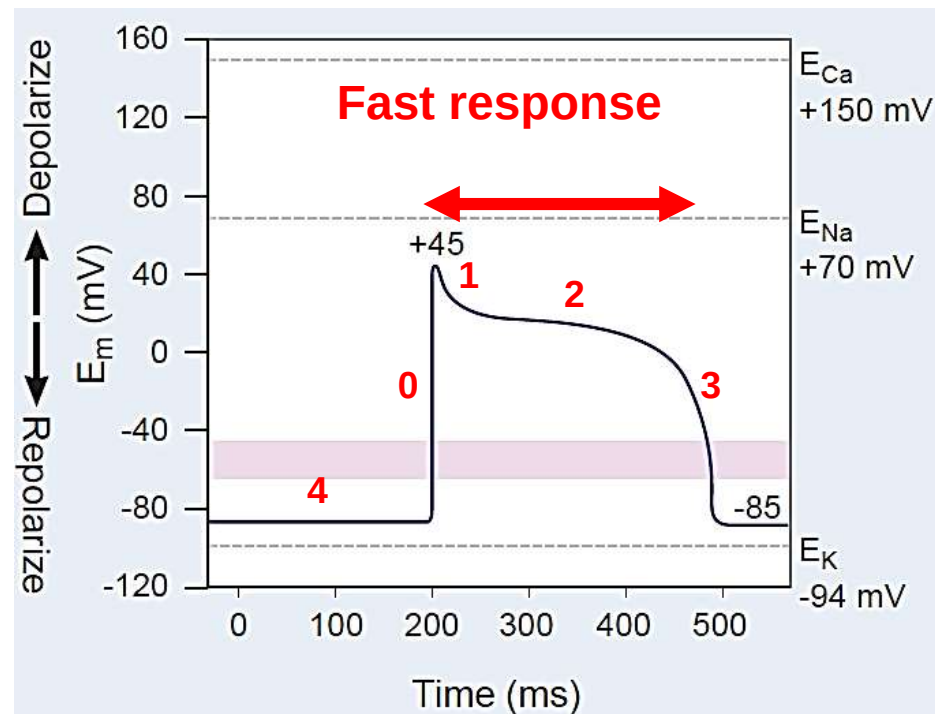


Cardiac action potentials: Slow & fast response

SA node cells



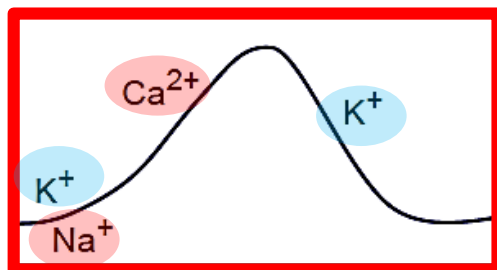
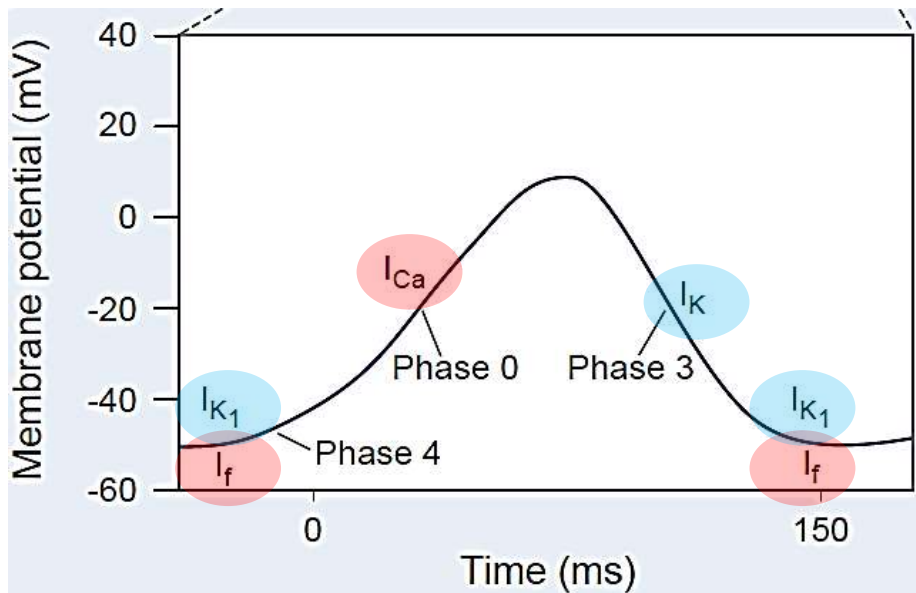
Ventricular muscle cells



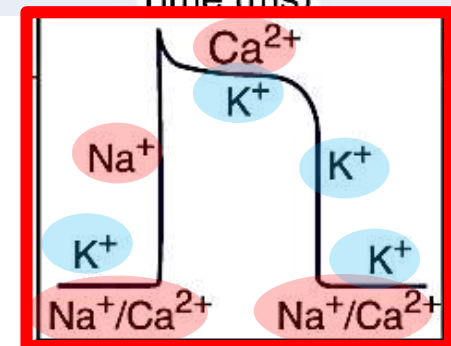
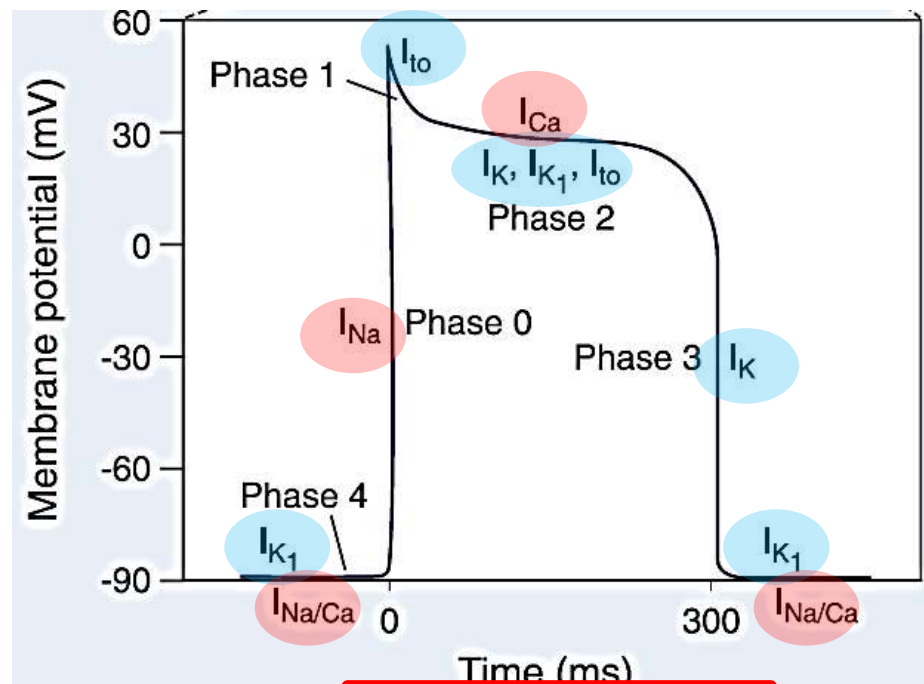
Cardiac action potentials are **longer** than those of nerve or skeletal muscles. Prolonged action potentials provide **sustained depolarization and contraction needed to empty the heart's chambers.**

Cardiac action potentials: SA nodal cells & Ventricular muscle cells

SA node cells



Ventricular muscle cells



Classification of arrhythmias

- **SA node arrhythmias**

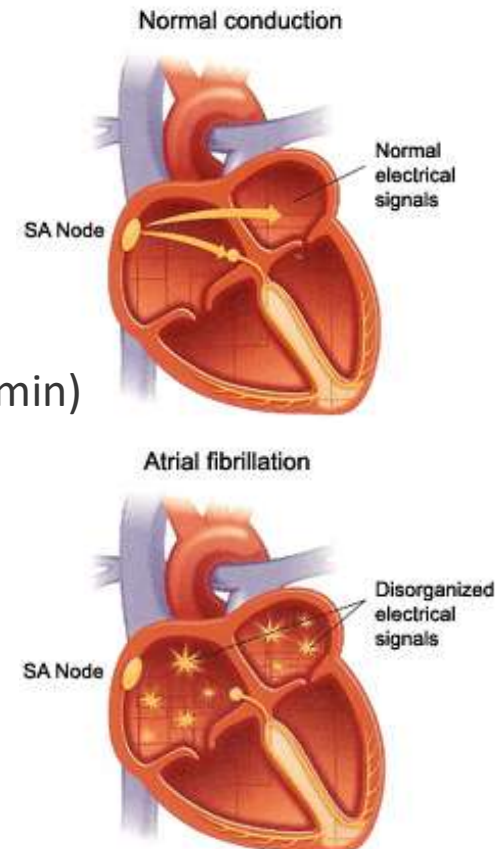
- **Sinus arrhythmia:** Irregular rhythm from SA node
- **Sinus bradycardia:** Regular rhythm, slow (<60 beats/min)
- **Sinus tachycardia:** Regular rhythm, fast (>100 beats/min)
- **Sinus arrest:** No rhythm

- **Atrial arrhythmias**

- **Premature atrial contractions (PACs)**
- **Atrial tachycardia**
- **Atrial flutter:** Regular rhythm, fast (>250-400 beats/min)
- **Atrial fibrillation:** Irregular rhythm, fast (>400-700 beats/min)

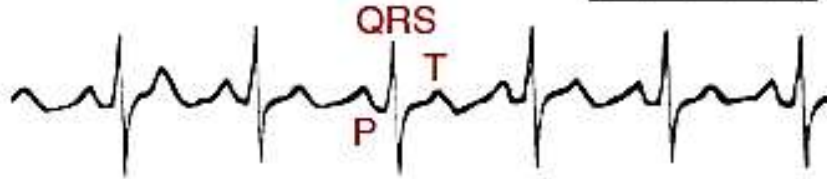
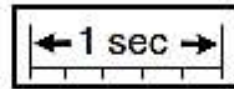
- **Ventricular arrhythmias**

- **Premature ventricular contractions (PVCs)**
- **Ventricular tachycardia**
- **Ventricular flutter**
- **Ventricular fibrillation**



Example of cardiac electrical disturbances

A Normal rhythm



B Premature ventricular beat



C Paroxysmal supraventricular tachycardia (PSVT)



D Atrial fibrillation



E Atrial flutter with variable AV conduction



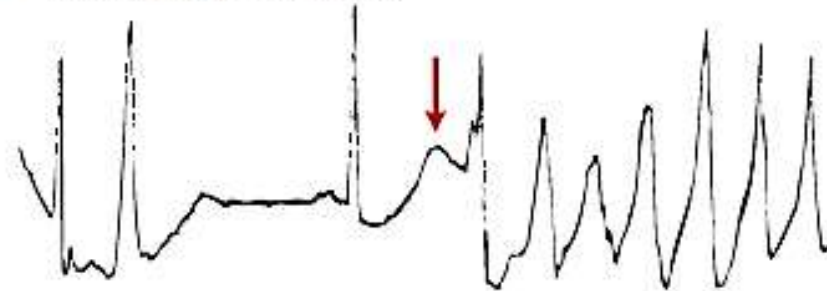
F Atrial flutter with 1:1 AV conduction



G Monomorphic ventricular tachycardia



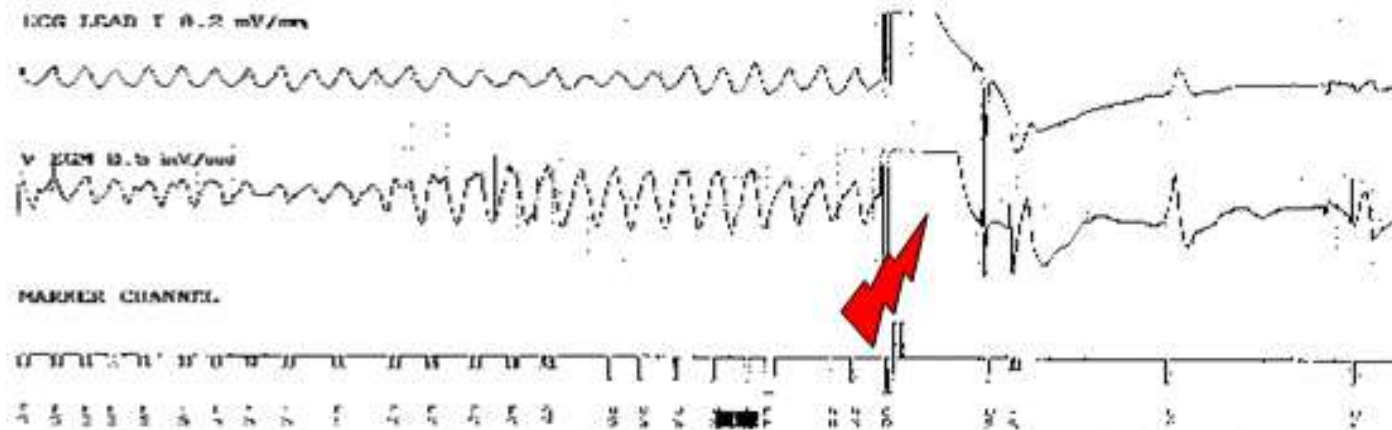
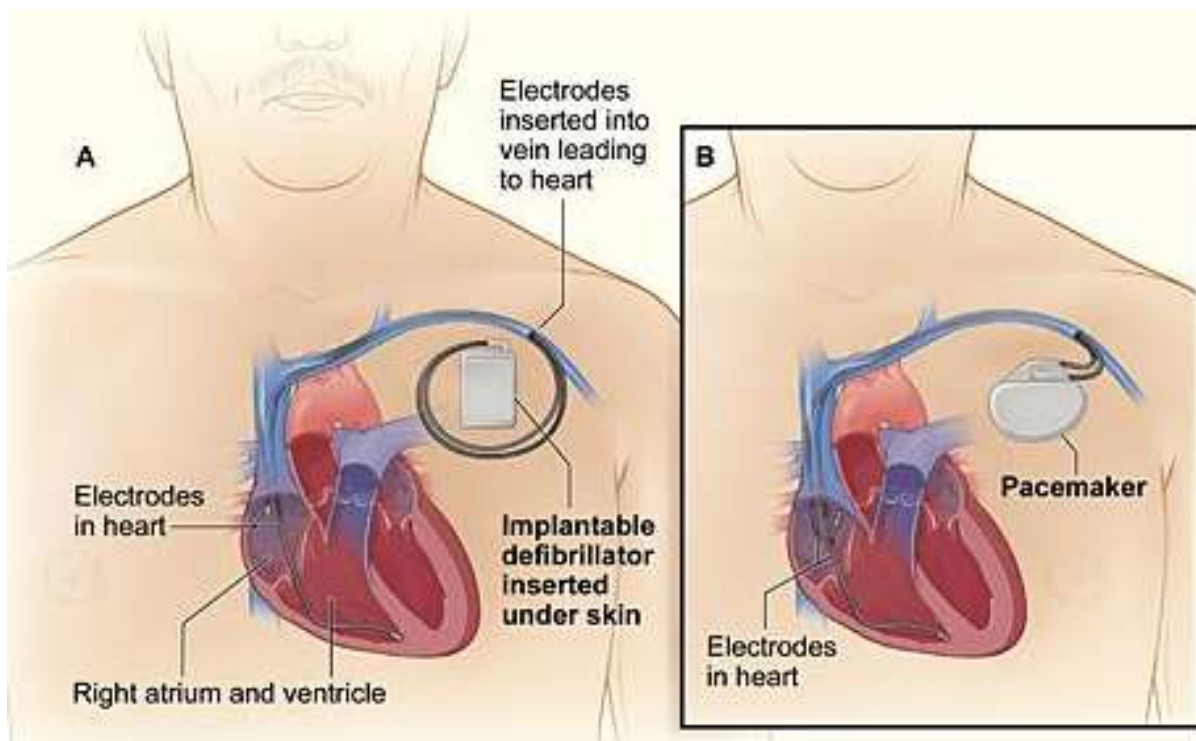
H Torsades de Pointes



I Ventricular fibrillation



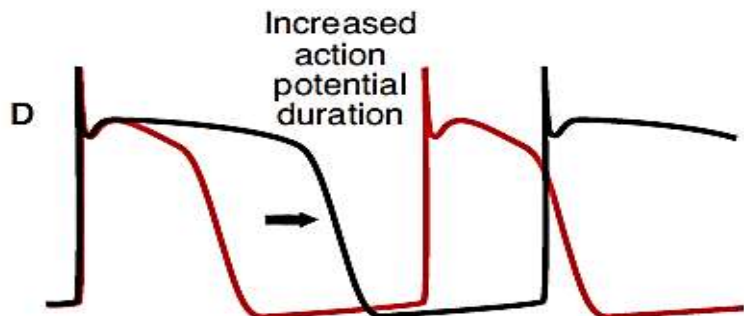
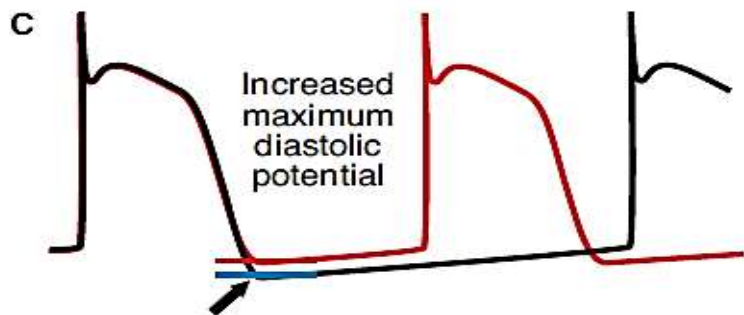
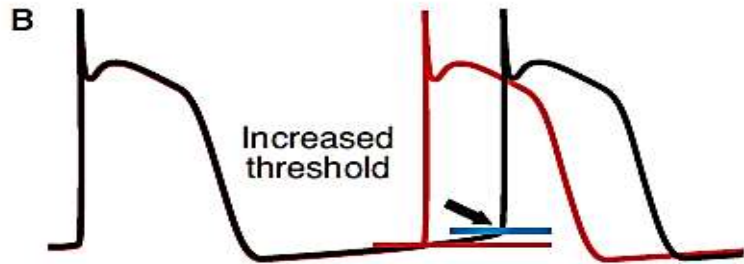
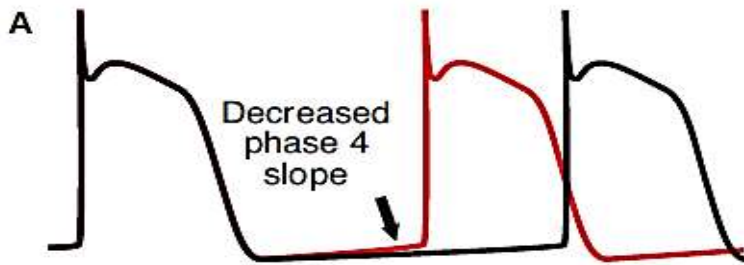
Implantable Cardioverter Defibrillator (ICD) & Pacemaker



Classification of antiarrhythmic drugs



General mechanisms of action of antiarrhythmic drugs



Drugs that affect cardiac rhythm act by altering:

↓ Rate of spontaneous AP

A Rate of phase 4 depolarization

Decrease slope

B Threshold potential

Increase

**C Maximum diastolic potential
(or resting membrane potential)**

More negative

D Action potential duration

Increase

Classes of antiarrhythmic drugs

Vaughan Williams's classification

Class	Examples	Mechanism
IA	<i>Quinidine, Procainamide, Disopyramide</i>	Sodium channel block (<u>intermediate</u> dissociation)
IB	<i>Lidocaine (E)</i> , Tocainide, Mexiletine, phenytoin	Sodium channel block (<u>fast</u> dissociation)
IC	<i>Flecainide (E)</i> , Encainide, <i>Propafenone (E)</i>	Sodium channel block (<u>slow</u> dissociation)
II	<i>Propranolol, esmolol, acebutolol</i>	β -adrenergic receptor antagonist
III	<i>Amiodarone (E)</i> , dronedarone, sotalol, ibutilide, dofetilide	Potassium channel block
IV	<i>Verapamil (E), diltiazem (E)</i>	Calcium channel block

(E): อยู่ในรายการบัญชียาหลักแห่งชาติ พ.ศ. 2567

Antiarrhythmic drugs unclassified in Vaughan Williams system

Drug	Use
<i>Adenosine (E)</i>	Supraventricular tachycardia
Magnesium sulfate (E)	Ventricular fibrillation
Adrenaline (epinephrine)	Cardiac arrest
Isoprenaline	Heart block
<i>Digoxin</i>	Rapid atrial fibrillation
Calcium chloride	Ventricular tachycardia due to hyperkalemia

(E): อยู่ในรายการบัญชียาหลักแห่งชาติ พ.ศ. 2567

Class IA: Quinidine

- **Mechanism:**

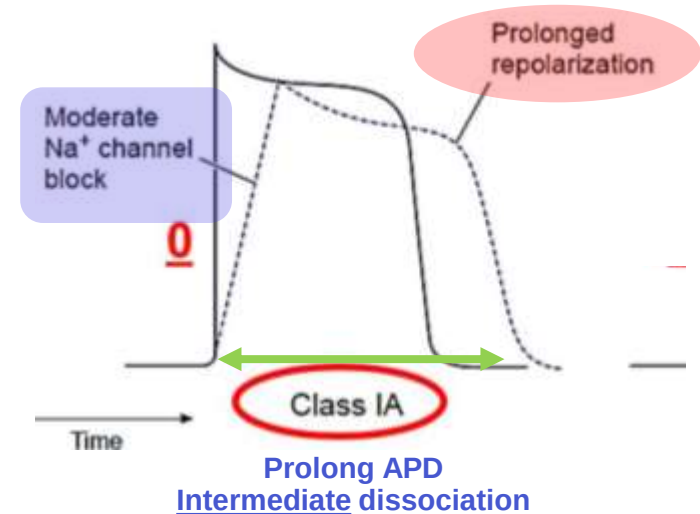
- Direct: **Inhibit Na^+ channels**
- Indirect: **Anticholinergic effect****
(**$\uparrow$ AV conduction, $\uparrow$ contractility**)

- **Action:**

- Depress pacemaker rate
- Reduce conduction & excitability of atria, His-Purkinje system, ventricle

- **Clinical uses:**

- Conversion of atrial flutter or fibrillation and maintenance of normal sinus rhythm
- PSVT
- Premature atrial/ventricular contractions



- **Pharmacokinetics:**

- **F** 70-80%
- **$T_{1/2}$** 5-12 h
- Protein binding 80-90%
- CYP450 (3A4, 2D6)
- Excretion: urine 20% (unchanged)

Class IA: Quinidine

- **Side effects:**
 - **GI (Common):** Diarrhea, nausea, vomiting
 - **Cinchonism:** Dizziness, tinnitus, headache
 - **Hypokalemia**
 - **Cardiovascular toxicity**
 - Increase AV conductance
 - Prolonged QT-interval, *torsade de pointes*
 - Hypotension
 - Quinidine syncope
 - Others: Thrombocytopenia, muscle weakness
- **Drug interaction:**
 - **digoxin (quinidine-induced digoxin toxicity)**
 - **warfarin**, verapamil, cimetidine, tricyclic antidepressant, neuromuscular blocker
- **Contraindications:**
 - Partial or complete AV block
 - Severe renal disease
 - Digitalis - induced arrhythmia
 - Myasthenia gravis
 - History of drug induced *Torsades de points*

Class IA: Procainamide (Pronestyl®)

- **Action:**

- Same as quinidine

- **Side effects:**

- Same as quinidine
- Fewer anticholinergic effects*
- Lupus-like syndrome (prolonged use)

- **Contraindications:**

- Same as quinidine
- Systemic lupus erythematosus (SLE)

- **Clinical uses:**

- PVC
- Life-threatening ventricular tachycardia
- Conversion of atrial flutter or fibrillation and maintenance of normal sinus rhythm

- **Pharmacokinetics:**

- Acetylation in liver to
N-acetylprocainamide (NAPA)
 - Active metabolite, $\uparrow T_{1/2}$
 - Na^+ block & K^+ block (III)

Lupus-like syndrome



Class IA: Disopyramide (Norpace®)

- **Action:**
 - Same as quinidine
- **Side effects:**
 - **More anticholinergic effects: GI, CVS**
- **Clinical uses:**
 - PVC
 - Ventricular tachycardia
 - Conversion of atrial flutter or fibrillation and maintenance of normal sinus rhythm
- **Contraindication:**
 - Same as quinidine

Class IB: Lidocaine (Xylocaine®)

- **Action:**

- Decrease depolarization & automaticity of ventricular cells

- **Side effects:**

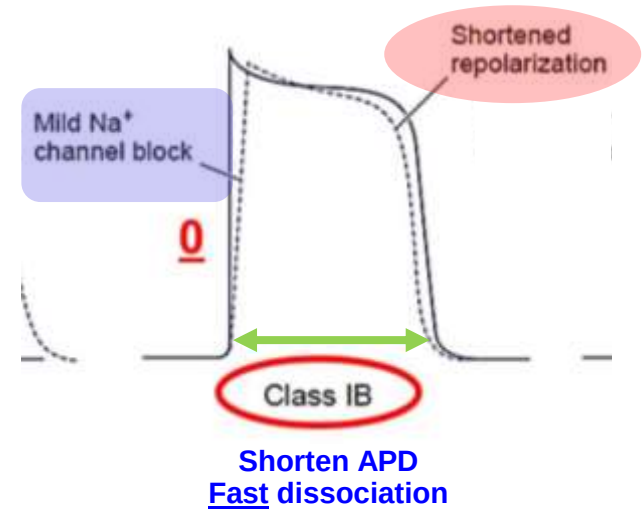
- **Less cardiotoxic, hypotension** (large doses)
- **CNS toxicity:** Confusion, twitching, seizure

- **Clinical uses:**

- **Drug of choice for acute ventricular tachycardia**
- Ventricular arrhythmia from cardiac surgery, MI, cardiac glycosides
- Rapid and safe by **IV route**
- **Not for routine use** due to increased mortality rate

- **Pharmacokinetic:**

- **Extensive first pass metabolism (IV, reduced dose in liver disease)**
- **Short $T_{1/2}$ (1-2 hr)**



Class IC: Flecainide (Tambocor[®])

- **Action:**

- Decrease depolarization & automaticity of ventricular cells
- **Proarrhythmic effects: Increase mortality (2.5x)****

- **Side effects:**

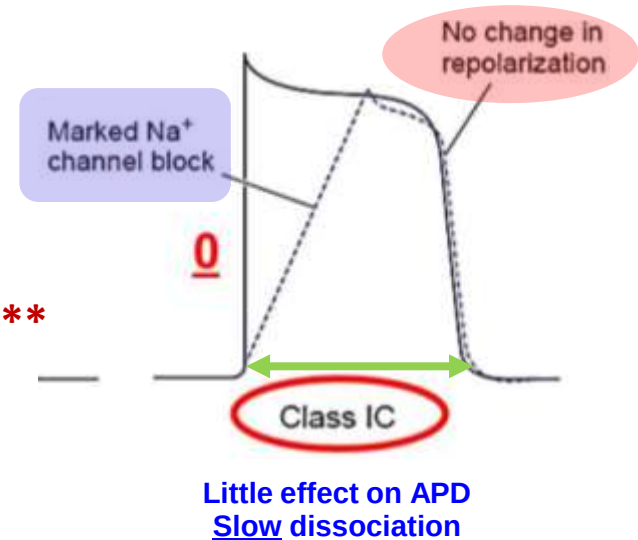
- Negative inotropy

- **Clinical uses:**

- Sustained ventricular tachycardia
- PSVT, paroxysmal atrial fibrillation unresponsive to other measures

- **Contraindications:**

- 2nd or 3rd degree heart block without a pacemaker, cardiogenic shock, recent MI, hypersensitivity



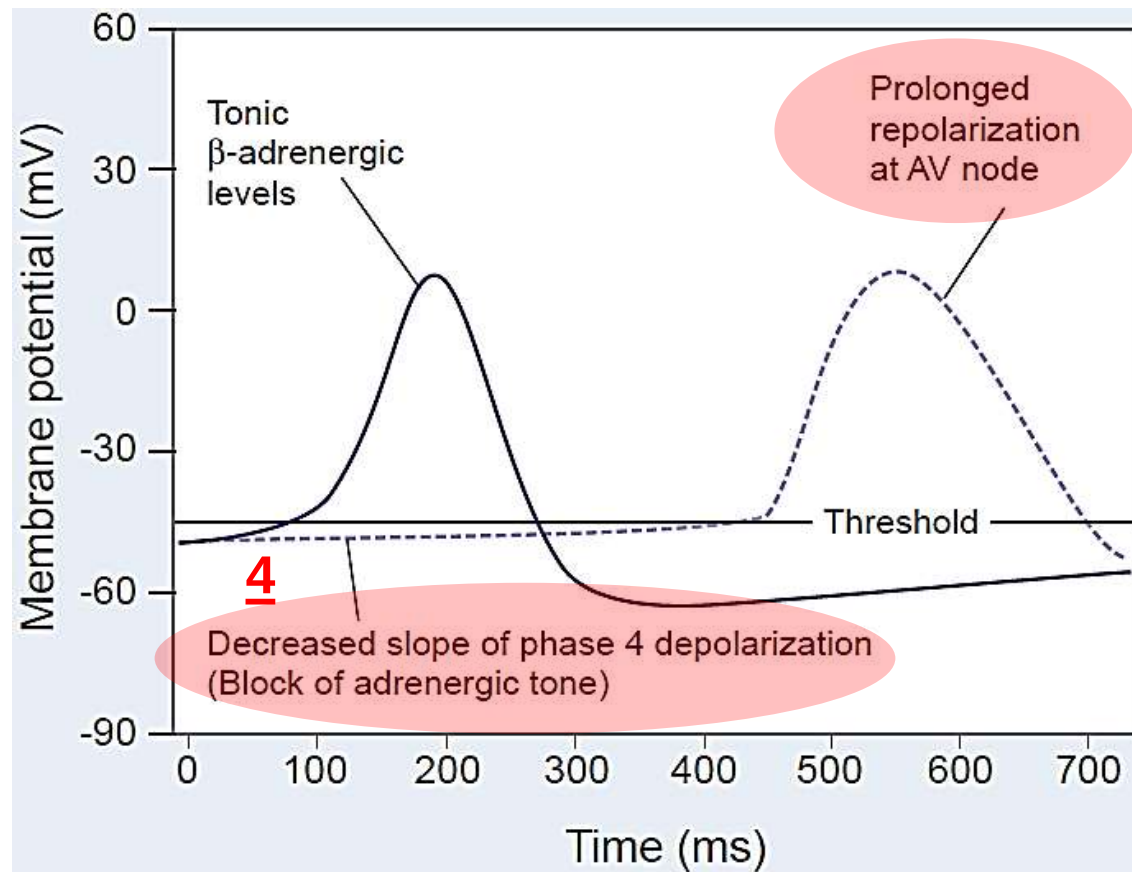
Class IC: Propafenone (Rytmonorm[®])

- **Action: ** Non-specific**
 - Structure similar to propranolol (**weak β -blocking action**)
 - Prolong repolarization (weak)
 - Calcium channel blocking action (weak)
- **Side effects:**
 - **GI (common): Metallic taste, constipation**
 - CVS: Proarrhythmias, negative inotropic effect
 - CNS: Dizziness, headache, visual changes
- **Clinical uses:**
 - Sustained ventricular tachycardia
 - PSVT, paroxysmal atrial fibrillation unresponsive to other measures
- **Pharmacokinetics:**
 - **Extensive first pass hepatic metabolism**

Class II Antiarrhythmic agents: β -adrenergic antagonists (β -blockers)

Effects on pacemaker cells action potentials

Phase 4 (Slow depolarization): Inward I_f current (mainly by Na^+)



Class II: β -blockers

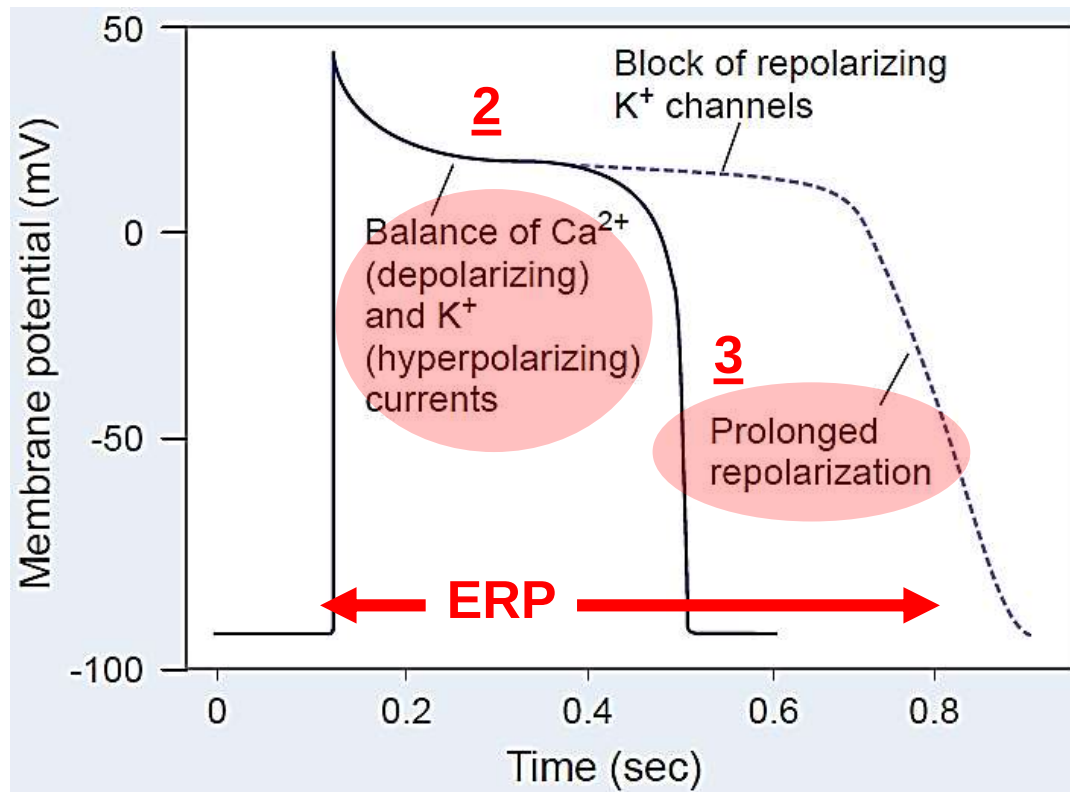
- **Action:**
 - Reduction of phase 4 depolarization
 - Increase effective refractory period (ERP) of AV node
- **Clinical uses:**
 - Supraventricular tachyarrhythmias, ventricular arrhythmia induced by catecholamines or digitalis
- **Drugs:**
 - **Propranolol**
 - Reduces the incidence of sudden arrhythmic death after myocardial infarction
 - **Acebutolol**
 - Premature ventricular beats
 - **Esmolol**
 - Very short-acting β -blocker used for **IV administration** in acute arrhythmias occur during surgery or emergency situations

Class III Antiarrhythmic agents: K⁺ channel blockers (Inhibitors of repolarization)

Effects on ventricular action potential

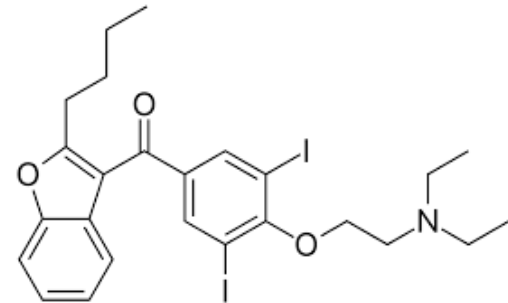
Phase 2 (Plateau): Balance between inward Ca²⁺ & outward K⁺ current

Phase 3 (Rapid repolarization): Decrease in inward Ca²⁺ current and large increase in outward K⁺ current



Class III: Amiodarone (Cordarone®)

- **Actions:**
 - **Block K⁺ channels (class III, main*)**
 - **Block Na⁺ channels (class I), β -receptors (class II), Ca²⁺ channels (class IV)**
- **Cardiac effects:**
 - Reduce inward Na⁺ current in cardiac cells
 - Prolongs refractory period in AV node
 - ↓ Excitability, conduction, velocity & automaticity in cardiac tissue
- **Extracardiac effects:**
 - **Peripheral vasodilation**
 - **Block peripheral conversion of T4 to T3**
- **Clinical uses:**
 - Recurrent ventricular fibrillation, unstable ventricular tachycardia
 - Atrial fibrillation, Supraventricular arrhythmias



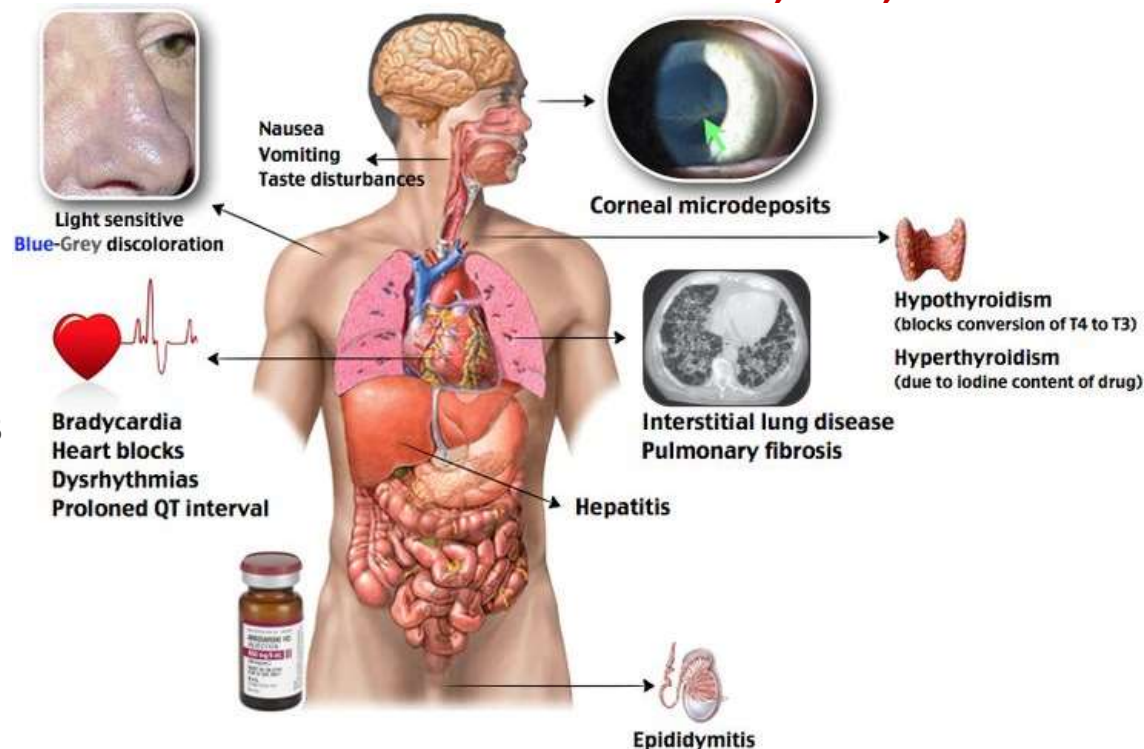
Class III: Amiodarone

Side effect ***

- Cardiac effects
 - ↓ AV- or SA-node function
 - ↓ Cardiac contractility
 - Hypotension
- Pulmonary
 - **Pneumonitis** leading to pulmonary fibrosis
- Thyroid
 - **Hyper / hypothyroidism**
- Hepatic
 - **Elevated liver enzymes**
- CNS
 - **Peripheral neuropathy**, headache, ataxia, tremors
- Other
 - **Corneal microdeposits**
 - Testicular dysfunction
 - **Skin discoloration**

Pharmacokinetic

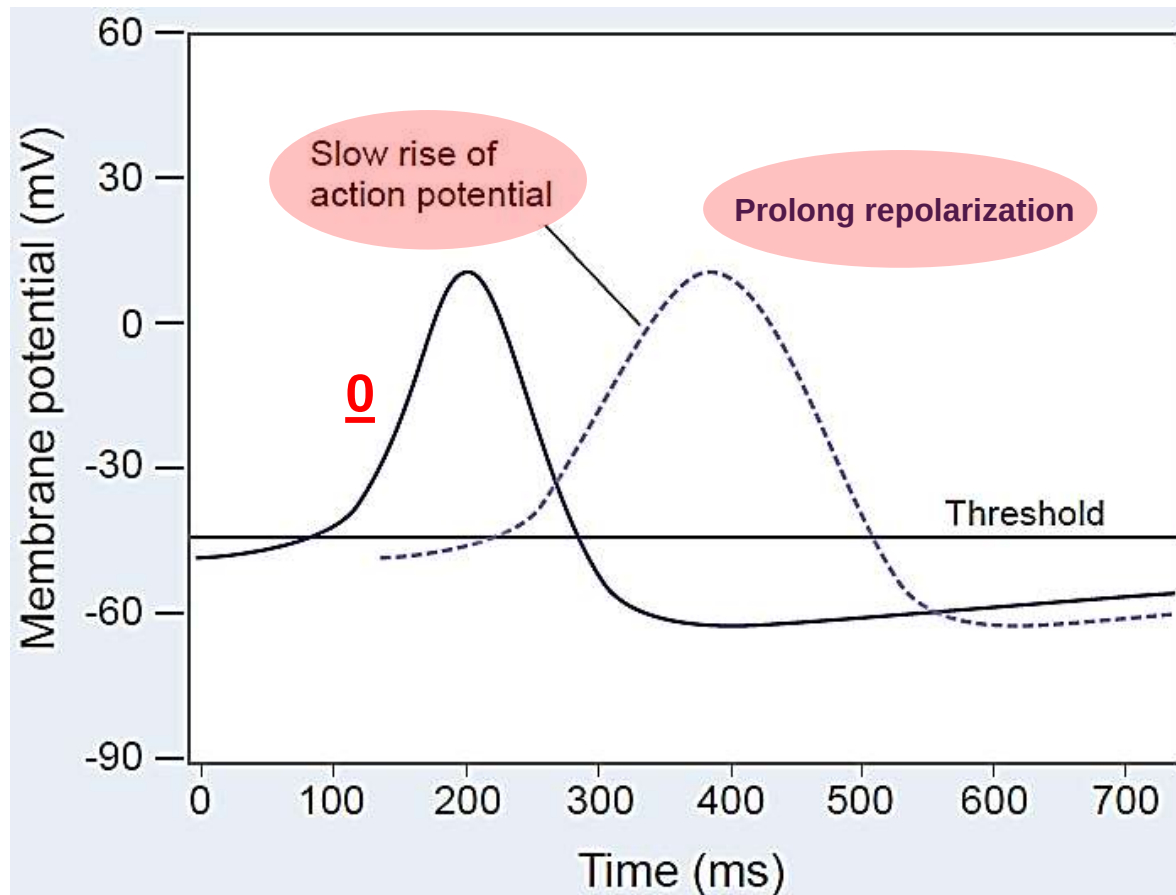
- Absorption: **Slow, variable**
- Metabolism
 - Desmethyl metabolite (active)
- **Long $T_{1/2}$, slow onset**
- **Inhibitors of CYP2D6, 2C9, 3A4**



Class IV Antiarrhythmic agents: Ca²⁺ channel blockers (CCBs)

Effects on pacemaker cells action potentials

Phase 0 (AP upstroke): Inward Ca²⁺ current (Ca²⁺ channels)



Class IV: Ca²⁺ channel blockers

- **Drugs:** Non-DHP CCBs (verapamil, diltiazem)
- **Actions:**
 - Cardiac effects:
 - Prolong AV nodal conduction time
 - Slowing AP upstroke -> slowed conduction velocity
 - Suppress both early and delayed afterdepolarization
 - Extracardiac effects:
 - Vasodilation
- **Indications:**
 - Supraventricular tachycardia
 - Reduce ventricular rate in atrial fibrillation and flutter
 - Rarely use in ventricular tachycardia
- **Precautions:**
 - AV block (excessive reduce conduction velocity)

Other: Adenosine (Adenocor®)

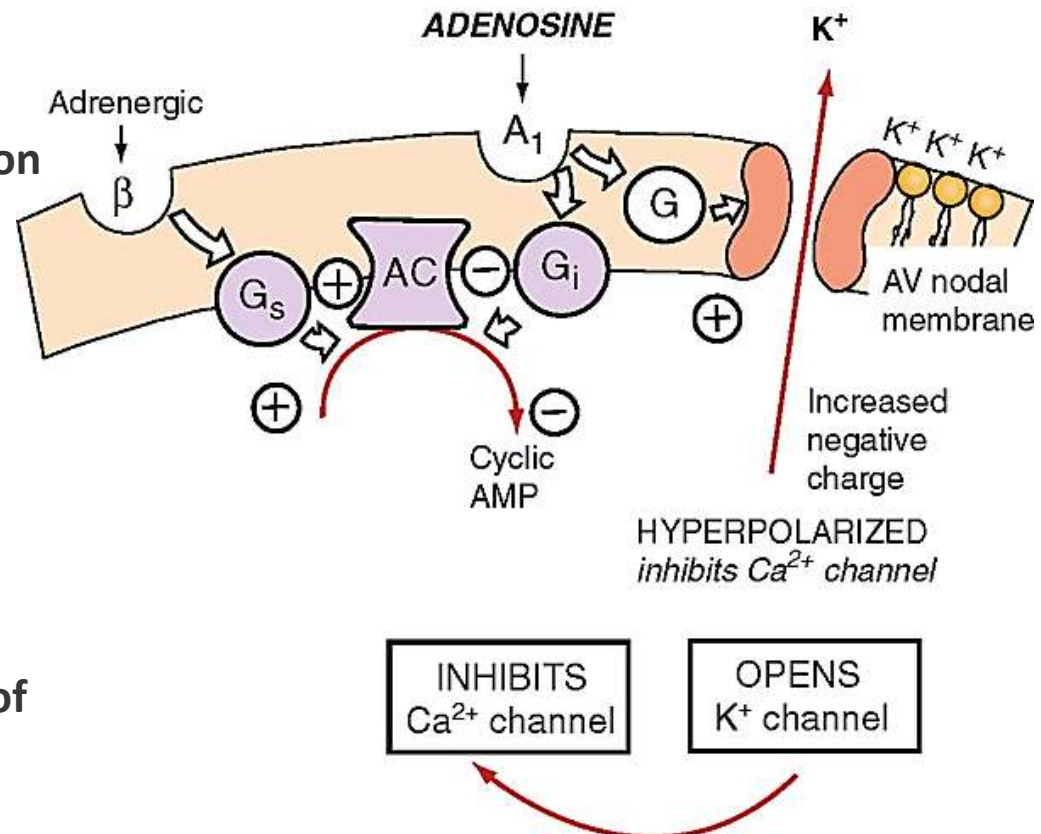
- **Mechanism:**
 - **Activate A_1 -receptor on cardiac tissues**
 - $\uparrow$ K^+ conductance
 - $\downarrow$ Ca^{2+} mediated conduction

- **Actions:**
 - Depress SA node, AV conduction
 - Terminate re-entry

- **Clinical uses:**
 - PSVT (rapid IV bolus)

- **Side effects:**
 - Dyspnea, flushing, heart block

- **Caution:**
 - Less effective in the presence of **adenosine receptor blockers (theophylline or caffeine)**



Other: Digitalis glycosides

- **Drug:**

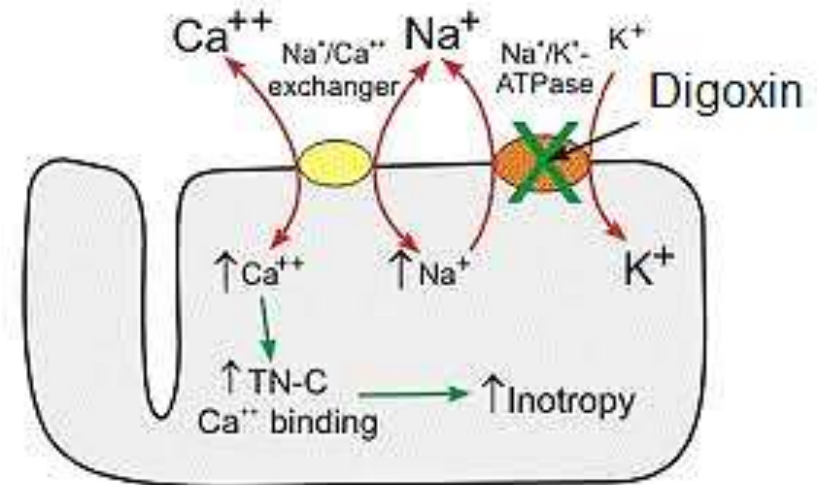
- **Digoxin (Inotropic drugs)**

- **Actions:**

- **Direct effect (inhibit $\text{Na}^+\text{-K}^+$ ATPase):** $\uparrow$ Contractility
- **Indirect vagomimetic effect:** Slows cardiac conduction & HR
 - $\uparrow$ vagal tone
 - $\downarrow$ central sympathetic outflow
 - $\downarrow$ myocardial response to β -ARs stimulation
 - $\downarrow$ RAAS activation

- **Clinical uses:**

- Used for atrial flutter/fibrillation

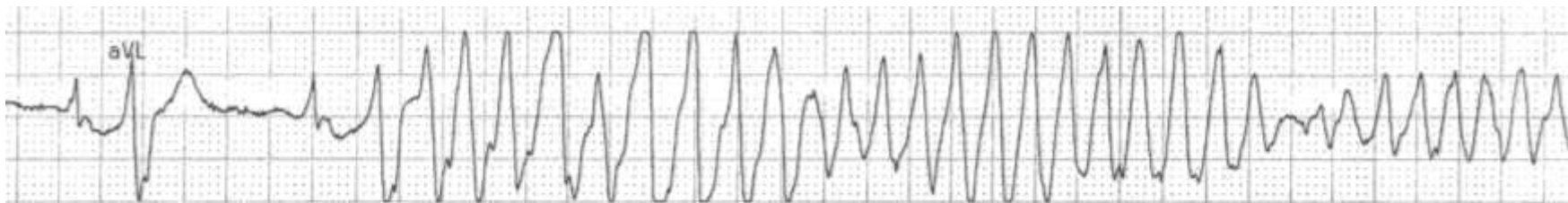
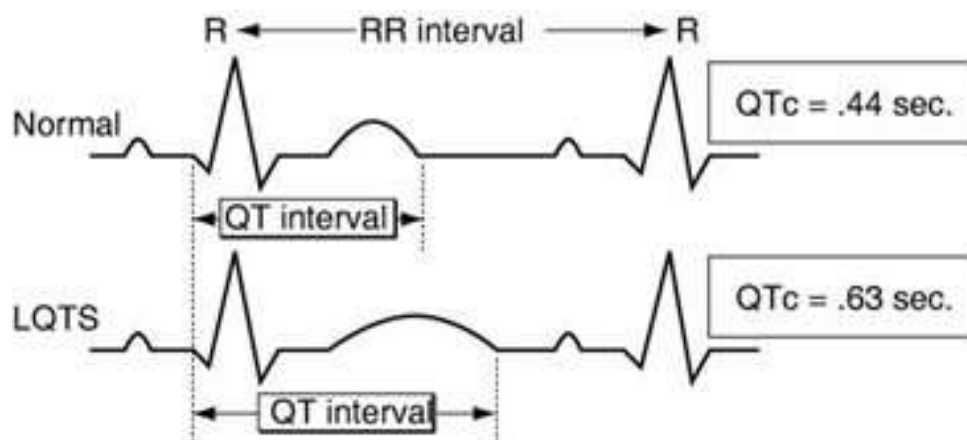


Drug-induced QT prolongation → *Torsades*



- **Long QT syndrome (LQTS)**

- Corrected QT interval (**QTc**) > 440 ms (Men), > 460 ms (Women)



Torsade de Pointes (TdP): Twisting of the point

QTc > 500 - > ↑ risk of TdP

Drug-induced QT prolongation

Drugs that may cause Torsade de Pointes

- **Antiarrhythmics:** disopyramide, procainamide, quinidine, amiodarone
- **Prokinetic agent:** cisapride, domperidone
- **Antibiotic:** macrolide (clarithromycin, erythromycin), trimethoprim-sulfamethoxazole, fluoroquinolone (moxifloxacin, gatifloxacin), voriconazole
- **Antipsychotic agents:** phenothiazine (chlorpromazine, thioridazine), tricyclic antidepressant, haloperidol, atypical antipsychotic (quetiapine, ziprasidone)
- **Miscellaneous:** digoxin, corticosteroid, diuretic (hydrochlorothiazide), quinine, chloroquine, tacrolimus, methadone

Risk factors for drug-induced Torsade de Pointes

- **Hypokalemia** ($K^+ < 3.5$ mEq/L) (e.g, thiazide)
- **Hypomagnesemia** ($Mg^{2+} < 1.5$ mEq/L)
- Bradycardia (HR < 60 bpm)
- Congestive heart failure (CHF)
- High drug concentrations
- Rapid rate of IV infusion with QT-prolonging drug
- Base-line QT prolongation

Principles in the clinical use of antiarrhythmic agents

- **Before initiation: Pretreatment evaluation**
 - **Eliminate the cause**
 - Homeostasis: Electrolyte imbalance (hypokalemia, hypomagnesemia)
 - Drug therapy & underlying disease: Hyperthyroidism, cardiac disease
 - **Make a firm diagnosis**
 - **Determine the baseline condition:** QTc interval
 - **Question the need for therapy:** Benefits & risks
- **After:**
 - **Drug selection:** Low side effects, proper dose & route, consider individual risk factors
 - **Concurrent drugs used**
 - **Risk factors:** Electrolytes (K^+ , Mg^{2+})
 - **Monitor QTc interval**

